

**Example 1.** Fit a straight line to the following data :

| Year $x$                  | 1961 | 1971 | 1981 | 1991 | 2001 |
|---------------------------|------|------|------|------|------|
| Production $y$<br>in tons | 8    | 10   | 12   | 10   | 16   |

and find the expected production in 2006.

**Soln.:-** The normal equations from least square method for a straight line  $y = ax + b$  is

$$a \sum_{i=1}^n x_i + nb = \sum_{i=1}^n y_i$$

$$a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i$$

Here  $n = 5$ . From the given data, we have, here we can consider  $x = 1, 2, 3, 4, 5$

| $x$                     | $y$                     | $xy$                         | $x^2$                     |
|-------------------------|-------------------------|------------------------------|---------------------------|
| 1                       | 8                       | 8                            | 1                         |
| 2                       | 10                      | 20                           | 4                         |
| 3                       | 12                      | 36                           | 9                         |
| 4                       | 10                      | 40                           | 16                        |
| 5                       | 16                      | 80                           | 25                        |
| $\sum_{i=1}^5 x_i = 15$ | $\sum_{i=1}^5 y_i = 56$ | $\sum_{i=1}^5 x_i y_i = 184$ | $\sum_{i=1}^5 x_i^2 = 55$ |

Therefore the normal equations are given by:

$$15a + 5b = 56$$

$$55a + 15b = 184$$

On solving above equations, we get

$$a = 1.6 \quad \text{and} \quad b = 6.4$$

Hence the best straight line is  $y = 1.6x + 6.4$ .

The expected production in 2006 is obtained by taking  $x = 5.5$ , we get  $y(2006) = 15.2$  tons.

**Example 2.** The results of measurements of electric resistance  $R$  of a copper bar at various temperatures  $t^{\circ}\text{C}$  are listed below :

|     |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|
| $t$ | 19 | 25 | 30 | 36 | 40 | 45 | 50 |
| $R$ | 76 | 77 | 79 | 80 | 82 | 83 | 85 |

Find a relation  $R = a + bt$  where  $a$  and  $b$  are constants to be determined.

**Soln.:-** Consider the normal equations of least square fit of a straight line i.e.

$$an + b \sum_{i=1}^n t_i = \sum_{i=1}^n R_i$$

$$a \sum_{i=1}^n t_i + b \sum_{i=1}^n t_i^2 = \sum_{i=1}^n t_i R_i$$

Here  $n = 7$ . From the given data, we have,

| $t$                      | $R$                      | $t^2$                       | $tR$                           |
|--------------------------|--------------------------|-----------------------------|--------------------------------|
| 19                       | 76                       | 361                         | 1444                           |
| 25                       | 77                       | 625                         | 1925                           |
| 30                       | 79                       | 900                         | 2370                           |
| 36                       | 80                       | 1296                        | 2880                           |
| 40                       | 82                       | 1600                        | 3280                           |
| 45                       | 83                       | 2025                        | 3735                           |
| 50                       | 85                       | 2500                        | 4250                           |
| $\sum_{i=1}^7 t_i = 245$ | $\sum_{i=1}^7 R_i = 562$ | $\sum_{i=1}^7 t_i^2 = 9307$ | $\sum_{i=1}^7 t_i R_i = 19884$ |

Therefore the normal equations are given by:

$$7a + 245b = 562$$

$$245a + 9307b = 19884$$

On solving above equations, we get

$$a = 70.0535 \quad \text{and} \quad b = 0.2923$$

Hence the best relation between  $t$  and  $R$  is  $R = 70.0535t + 0.2923$ .

**Example 3.** If  $P$  is the pull required to lift a load  $W$  by means of a pulley block, find a linear law of the form  $P = mW + C$ , connecting  $P$  and  $W$ , using the following data

|             |    |    |     |     |
|-------------|----|----|-----|-----|
| $P$ (in Kg) | 12 | 15 | 21  | 25  |
| $W$ (in Kg) | 50 | 70 | 100 | 120 |

Use least square method to compute  $P$  when  $W = 150$ kg.

**Soln.:-** The normal equations from least square method for a straight line  $P = mW + C$  is

$$m \sum_{i=1}^n W_i + nC = \sum_{i=1}^n P_i$$

$$m \sum_{i=1}^n W_i^2 + C \sum_{i=1}^n W_i = \sum_{i=1}^n W_i P_i$$

Here  $n = 4$ . From the given data, we have,

| $P$                     | $W$                      | $W^2$                          | $WP$                        |
|-------------------------|--------------------------|--------------------------------|-----------------------------|
| 12                      | 50                       | 144                            | 600                         |
| 15                      | 70                       | 225                            | 1050                        |
| 21                      | 100                      | 441                            | 2100                        |
| 25                      | 120                      | 625                            | 3000                        |
| $\sum_{i=1}^4 P_i = 73$ | $\sum_{i=1}^4 y_i = 340$ | $\sum_{i=1}^4 x_i y_i = 31800$ | $\sum_{i=1}^4 x_i^2 = 6750$ |

Therefore the normal equations are given by:

$$340m + 4C = 73$$

$$31800m + 340C = 6750$$

On solving above equations, we get

$$m = 0.1879 \quad \text{and} \quad C = 2.2759$$

Hence the best relation between  $P$  and  $W$  is  $P = 0.1879W + 2.2759$ .

The value of  $P$  when  $W = 150$  is approximately 30.46 Kg.

## Fitting a Parabola (Quadratic Function)

Let  $y = f(x) = ax^2 + bx + c$  be the parabola to be fitted to the given set of  $n$  data points  $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$ .

The error between the data points and the parabola is

$$e_i = y_i - f(x_i) = y_i - (ax_i^2 + bx_i + c), \quad i = 1, 2, 3, \dots, n$$

The sum of the squares of the errors is

$$E = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2.$$

Now by the method of least square, for the curve of best fit,  $E$  is minimum

That is,

$$\frac{\partial E}{\partial a} = \frac{\partial E}{\partial b} = \frac{\partial E}{\partial c} = 0.$$

From the above equation, we get normal equations as:

$$\begin{aligned} a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i + cn &= \sum_{i=1}^n y_i \\ a \sum_{i=1}^n x_i^3 + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i \\ a \sum_{i=1}^n x_i^4 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n x_i^2 y_i \end{aligned}$$

After solving above normal equations we get values of  $a, b$  and  $c$ .

**Example 1.** Fit a parabola  $y = ax^2 + bx + c$  to the given data

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | 10 | 12 | 15 | 20 | 23 |
| $y$ | 14 | 17 | 23 | 21 | 25 |

**Soln.:** Let the parabola of best fit be given by

$$y = ax^2 + bx + c$$

Normal equations are given by:

$$\begin{aligned} a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i + cn &= \sum_{i=1}^n y_i \\ a \sum_{i=1}^n x_i^3 + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i \\ a \sum_{i=1}^n x_i^4 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n x_i^2 y_i \end{aligned}$$

| $x$                        | $y$                         | $xy$                             | $x^2$                          | $x^2y$                              | $x^3$                           | $x^4$                            |
|----------------------------|-----------------------------|----------------------------------|--------------------------------|-------------------------------------|---------------------------------|----------------------------------|
| 10                         | 14                          | 140                              | 100                            | 1400                                | 1000                            | 10000                            |
| 12                         | 17                          | 204                              | 144                            | 2448                                | 1728                            | 20736                            |
| 15                         | 23                          | 345                              | 225                            | 5175                                | 3375                            | 50625                            |
| 20                         | 21                          | 420                              | 400                            | 8400                                | 8000                            | 160000                           |
| 23                         | 25                          | 575                              | 529                            | 13225                               | 12167                           | 279841                           |
| $\sum_{i=1}^n x_i$<br>= 80 | $\sum_{i=1}^n y_i$<br>= 100 | $\sum_{i=1}^n x_i y_i$<br>= 1684 | $\sum_{i=1}^n x_i^2$<br>= 1398 | $\sum_{i=1}^n x_i^2 y_i$<br>= 30648 | $\sum_{i=1}^n x_i^3$<br>= 26270 | $\sum_{i=1}^n x_i^4$<br>= 521202 |

Substituting these values into normal equations, we get

$$\begin{aligned} 1398a + 80b + 5c &= 100 \\ 26270a + 1398b + 80c &= 1684 \\ 521202a + 26270b + 1398c &= 30648 \end{aligned}$$

After solving above linear systems, we get  $a = -0.0695, b = 3.0099, c = -8.7279$ .

$\therefore$  The parabola of best fit is  $y = -0.0695x^2 + 3.0099x - 8.7279$ .

**Example 2.** Fit a parabola of the form  $y = a + bx + cx^2$  which fit most closely with the observations:

| $x$ | -3   | -2   | -1   | 0    | 1    | 2    | 3    |
|-----|------|------|------|------|------|------|------|
| $y$ | 4.63 | 2.11 | 0.67 | 0.09 | 0.63 | 2.15 | 4.58 |

Evaluate  $y$  at  $x = 0.5$  and  $x = 1.5$

**Soln.:-** Let the parabola of best fit be given by

$$y = a + bx + cx^2$$

Normal equations are given by:

$$\begin{aligned} an + b \sum_{i=1}^n x_i + c \sum_{i=1}^n x_i^2 &= \sum_{i=1}^n y_i \\ a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i^3 &= \sum_{i=1}^n x_i y_i \\ a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i^3 + c \sum_{i=1}^n x_i^4 &= \sum_{i=1}^n x_i^2 y_i \end{aligned}$$

| $x$                    | $y$                        | $xy$                           | $x^2$                     | $x^2y$                            | $x^3$                    | $x^4$                      |
|------------------------|----------------------------|--------------------------------|---------------------------|-----------------------------------|--------------------------|----------------------------|
| -3                     | 4.63                       | -13.89                         | 9                         | 41.67                             | -27                      | 81                         |
| -2                     | 2.11                       | -4.22                          | 4                         | 8.44                              | -8                       | 16                         |
| -1                     | 0.67                       | -0.67                          | 1                         | 0.67                              | -1                       | 1                          |
| 0                      | 0.09                       | 0                              | 0                         | 0                                 | 0                        | 0                          |
| 1                      | 0.63                       | 0.63                           | 1                         | 0.63                              | 1                        | 1                          |
| 2                      | 2.15                       | 4.3                            | 4                         | 8.6                               | 8                        | 16                         |
| 3                      | 4.58                       | 13.74                          | 9                         | 41.22                             | 27                       | 81                         |
| $\sum_{i=1}^n x_i = 0$ | $\sum_{i=1}^n y_i = 14.86$ | $\sum_{i=1}^n x_i y_i = -0.11$ | $\sum_{i=1}^n x_i^2 = 28$ | $\sum_{i=1}^n x_i^2 y_i = 101.23$ | $\sum_{i=1}^n x_i^3 = 0$ | $\sum_{i=1}^n x_i^4 = 196$ |

Substituting these values into normal equations, we get

$$7a + 0b + 28c = 14.86 \Rightarrow a + 4c = 2.1229$$

$$0a + 28b + 0c = -0.11 \Rightarrow b = -0.0039$$

$$28a + 0b + 196c = 101.23 \Rightarrow a + 7c = 3.6154$$

After solving above linear systems, we get  $a = 0.1329, b = -0.0039, c = 0.4975$

$\therefore$  The parabola of best fit is  $y = 0.1329 - 0.0039x + 0.4975x^2$ .

$$\therefore y(0.5) = 0.2553 \quad \text{and} \quad y(1.5) = 1.2463.$$